

Machine Learning and Data Mining (ST443)

Zoltán Szabó

LSE

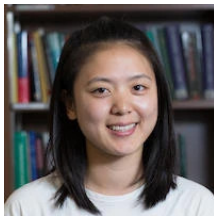


- Team.
- Reference & authors.
- ML hints.
- Admin: assessment, genAI policy.
- High-level idea: prototype tasks.
- (Linear) algebra: recap.

Team

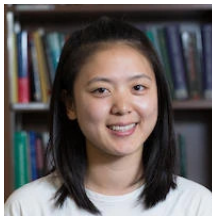
Team [notation: [link](#)]

- Seminar: [Phil Chan](#), [Kaifang Zhou](#).



Team [notation: [link](#)]

- Seminar: [Phil Chan](#), [Kaifang Zhou](#).



- Lecture: [Zoltan Szabo](#)



- ① Phil (3 groups):
 - guest lecturer,
 - class teaching: Programming for Data Science (ST101), Probability and Distribution Theory (ST206), Statistical Inference (ST216), Regression and Generalised Linear Models (ST300), Machine Learning (ST310), Artificial Intelligence (ST311), ML and Data Mining (ST443), RL (ST455).

Introduction: Phil, Kaifang

① Phil (3 groups):

- guest lecturer,
- class teaching: Programming for Data Science (ST101), Probability and Distribution Theory (ST206), Statistical Inference (ST216), Regression and Generalised Linear Models (ST300), Machine Learning (ST310), Artificial Intelligence (ST311), ML and Data Mining (ST443), RL (ST455).

② Kaifang (1 group):

- LSE Fellow,
- class teaching: Elementary Statistical Theory (ST102), Quantitative Methods (Statistics; ST107), Sample Surveys and Experiments (ST205), Stochastic Processes (ST302), Stochastic Simulation - computer workshop (ST303), ML and Data Mining (ST443), Managing and Visualising Data (ST445), Distributed Computing for Big Data (ST446).
- research interests: ranking algorithms, choice models.

Introduction: Zoltan

- ① background: applied maths, a bit of CS,
- ② research interests: stat.ML → information theoretical estimators, kernel techniques, scalable computation & their applications

Introduction: Zoltan

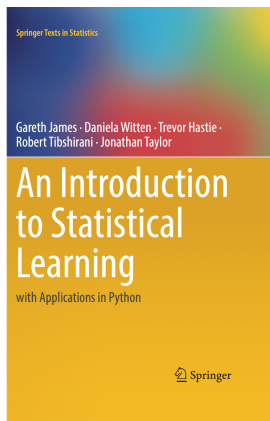
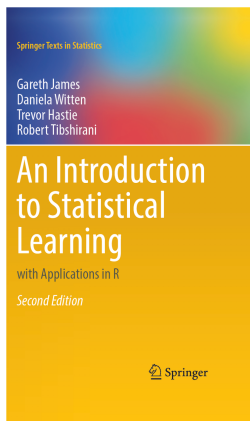
- ① background: applied maths, a bit of CS,
- ② research interests: stat.ML → **information theoretical estimators**, kernel techniques, scalable computation & their applications,
- ③ hobby: nature; canoeing, photography, ping-pong.



References & authors

Ref-1: backbone; minor branch outs when need arises

Gareth James, Daniela Witten, Trevor Hastie, Robert Tibshirani, Jonathan Taylor (Python). **An Introduction to Statistical Learning with Applications in R/Python**, Springer, 2021 (R) and 2023 (Python) =: ISLR/ISLP.



A few words about the authors

Gareth James:



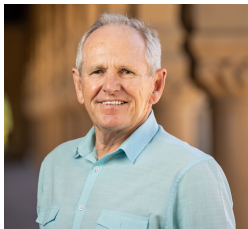
- Dean of UCLA Anderson School of Management.
- research:
 - statistics, data science, functional data analysis, marketing.
 - **Google Scholar**: 47,607 citations.
- his PhD advisor (1998): Trevor Hastie.

Daniela Witten:



- Professor of Statistics & of Biostatistics @ University of Washington.
- research:
 - statistics, machine learning; highD statistics, selective inference, multiview data; genomics, neuroscience, microbial ecology, pathology.
 - [Google Scholar](#): 68,873 citations.

Trevor Hastie:



- Professor of Statistics & of Biomedical Data Science @ Stanford University.
- research:
 - statistical learning and modelling, data mining, machine learning.
 - [Google Scholar](#): 468,975 citations.

Robert Tibshirani:

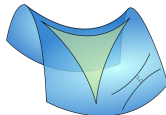


- Professor of Biomedical Data Science & of Statistics @ Stanford University.
- research:
 - statistics, data science, machine learning
 - [Google Scholar](#): 633,635 citations.

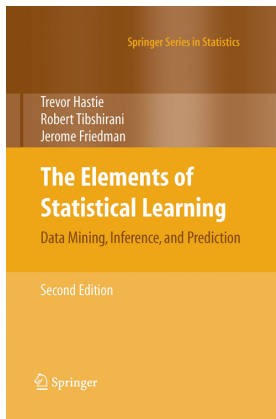
Jonathan Taylor:



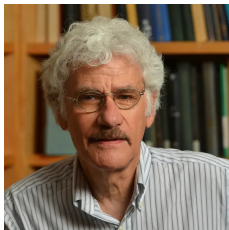
- Professor of Statistics @ Stanford University.
- research:
 - Gaussian processes, stochastic processes, differential geometry; neuroimaging, HIV protein sequence analysis.
 - [Google Scholar](#): 75,781 citations.



Trevor Hastie, Robert Tibshirani, Jerome Friedman. **Elements of Statistical Learning**, Springer, 2009.



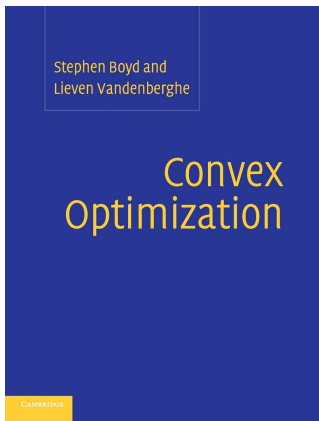
Jerome H. Friedman:



- Professor Emeritus of Statistics @ Stanford University.
- research:
 - statistics, data mining.
 - [Google Scholar](#): 395,023 citations.

Ref-3: optimization, duality

Stephen Boyd and Lieven Vandenberghe. **Convex Optimization**, Cambridge University Press, 2004.



Stephen Boyd:



- Professor of Electrical Engineering, Computer Science, and Management Science @ Stanford University.
- research:
 - optimization, control, signal processing, AI, finance.
 - [Google Scholar](#): 300,921 citations.

Lieven Vandenberghe:



- Professor @ Electrical and Computer Engineering Department & Department of Mathematics, University of California, Los Angeles.
- research:
 - optimization, systems and control, signal processing.
 - [Google Scholar](#): 115,968 citations.

Ref-4: non-smooth optimization

- Neal Parikh and Stephen Boyd. **Proximal Algorithms**. Foundations and Trends in Optimization, volume 1, issue 3, pp. 127-239, 2014.

- Authors continued (**Neal Parikh**):



- Adjunct Associate Professor @ School of International & Public Affairs, Columbia University.
- research:
 - machine learning, convex optimization, statistics, technology policy, AI ethics.
 - **Google Scholar**: 37,515 citations.

ML hints

Typesetting: $\text{\LaTeX}$

- high-quality typesetting system ($\text{\TeX}$) with macro packages ($\text{\LaTeX}$),
- creator:
 - The Art of Computer Programming (book, 7 volumes): publisher with poor typesetting $\Rightarrow$ Donald Knuth ($\text{\TeX}$, '78),
 - version number: 3.141592653 (in 2021) $\rightarrow \pi$,
 - coding errors in $\text{\TeX}$: $\$2.56 = 256$ cents, $\times 2$.



- Leslie Lamport ($\text{\LaTeX}$, '84).

$\text{\TeX}$ = high-quality article/book/TR/poster/... with minimal effort

- gold standard in ML,
- cross-platform: Windows, MacOS, Linux,
- CTAN = packages, ...
- collaborative editing: Overleaf ($\leftrightarrow$ Git),
- style: WYSIWYG instead of (all)WYSIWYG,

what you want
what you get
- 1/2h intro to L^AT_EX.

Top ML venues $\Rightarrow$ pick up the writing 'etiquette'

- Terminology:

AI $\approx$ ML $\approx$ data science $\approx$ applied maths $\approx$ data-centric engineering

Top ML venues $\Rightarrow$ pick up the writing 'etiquette'

- Terminology:

AI $\approx$ ML $\approx$ data science $\approx$ applied maths $\approx$ data-centric engineering

- Mathwise:

$\underbrace{\text{statistics}}_{\text{guarantees}}, \underbrace{\text{optimization}}_{\text{computation}}, \underbrace{\text{functional analysis}}_{\text{function spaces}}$

Top ML venues $\Rightarrow$ pick up the writing 'etiquette'

- Terminology:

AI $\approx$ ML $\approx$ data science $\approx$ applied maths $\approx$ data-centric engineering

- Mathwise:

$\underbrace{\text{statistics}}_{\text{guarantees}}, \underbrace{\text{optimization}}_{\text{computation}}, \underbrace{\text{functional analysis}}_{\text{function spaces}}$

- Top-tier **conferences** [acceptance $\sim 20 - 25\%$]:

- Conference on Neural Information Processing Systems (**NeurIPS**),
- International Conference on Machine Learning (**ICML**),
- International Conference on Artificial Intelligence and Statistics (**AISTATS**),
- Conference on Uncertainty in Artificial Intelligence (**UAI**),
- Conference on Learning Theory (**COLT**).

Top ML venues $\Rightarrow$ pick up the writing 'etiquette'

- Terminology:

AI $\approx$ ML $\approx$ data science $\approx$ applied maths $\approx$ data-centric engineering

- Mathwise:

statistics, optimization, functional analysis
guarantees computation function spaces

- Top-tier **conferences** [acceptance $\sim 20 - 25\%$]:

- Conference on Neural Information Processing Systems (**NeurIPS**),
- International Conference on Machine Learning (**ICML**),
- International Conference on Artificial Intelligence and Statistics (**AISTATS**),
- Conference on Uncertainty in Artificial Intelligence (**UAI**),
- Conference on Learning Theory (**COLT**).

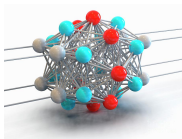
- Top **journal**:

- Journal of Machine Learning Research (**JMLR**).

Top conferences+

Great 'app-specific' conferences :

- NLP:
 - Empirical Methods in Natural Language Processing (EMNLP).
- deep & representation learning:
 - International Conference on Learning Representations (ICLR).
- computer vision:
 - Conference on Computer Vision and Pattern Recognition (CVPR),
 - International Conference on Computer Vision (ICCV).



- conference-oriented! (vs stats),
- extension to journal (+theory/app),
- open (typically),
- reproducibility: more & more emphasis.

History:

- open-access repo of preprints,
- hosted by Cornell University,
- launched in 1991 (Paul Ginsparg; T_EX made it possible),

History:

- covered areas: **mathematics**, **computer science**, physics, quantitative biology, quantitative finance, **statistics**, electrical engineering and systems science, economics.

History:

- covered areas: [mathematics](#), [computer science](#), physics, quantitative biology, quantitative finance, [statistics](#), electrical engineering and systems science, economics.
- [submission rate](#) (2025): 284,486 articles/year; [further stats](#).

- Categories [1st approximation]:
 - stat.ML: statistical machine learning ([stat](#)),
 - cs.LG: learning ([CS](#)),
 - math.OC: optimization ([maths](#)).
- Description of sub-categories in statistics and CS, classification:
 - [statistics](#), [CS](#),
 - [ML classification guide](#).

- Categories [1st approximation]:
 - stat.ML: statistical machine learning (**stat**),
 - cs.LG: learning (**CS**),
 - math.OC: optimization (**maths**).
- Description of sub-categories in statistics and CS, classification:
 - **statistics**, **CS**,
 - **ML classification guide**.
- Each paper has 1 primary and typically 1-2 secondary categories.
- In addition: **MSC** (=maths) and **ACM** (=CS) classes.

- ML:
 - [ML-News](#),
 - [UAI](#).
- Statistics: [allstat](#).
- ML + neuroscience: [Connectionists](#).
- Academic social networking: [LinkedIn](#).
- Computing job announcements: [CRA](#).
- Jobs in the UK: [jobs.ac.uk](#).

Programming languages

- Python:
 - dominant in data science.
 - general-purpose language $\xrightarrow{\text{specialized}}$ ML.
 - Good book: Mark Lutz. *Learning Python*, 6th Edition. O'Reilly Media, Inc., 2025.
- R:
 - widely-used in statistics.

@ LSE

Digital Skills Lab: pre-sessional courses (Python, R, L^AT_EX).

- : Windows, MacOS, Linux, else.



- : Windows, MacOS, Linux, else.



- Linux:
 - idea: user freedom $\Rightarrow$ you can own your computer.

- **?**: Windows, MacOS, Linux, else.



- **Linux:**
 - idea: user freedom $\Rightarrow$ you can own your computer.
 - free $\Rightarrow$
 - versatile, sustainable, secure, privacy-respecting,
 - runs 'everywhere': laptops/desktops, mobiles, paper-like tablets, research computing (cluster), ...

- **?**: Windows, MacOS, Linux, else.



- Linux:
 - idea: user freedom $\Rightarrow$ you can own your computer.
 - free $\Rightarrow$
 - versatile, sustainable, secure, privacy-respecting,
 - runs 'everywhere': laptops/desktops, mobiles, paper-like tablets, research computing (cluster), ...
- For the curious-minded:
 - [Getting Started with Linux](#) mini-course: slides (Sept. 25).



Linus Torvalds: also author of **Git**,

- distributed version control system,
- standard in collaborative coding efforts,
- developed for the Linux kernel ('05),
- # of lines in the code of Linux kernel: **40+ million** (2025).



OpenReview

- created by **Andrew McCallum**'s lab (Univ. of Massachusetts Amherst),
- aim: open scientific communication, discussion,
- widely-used for ML conferences,
- roles: author $\rightarrow$ reviewer $\rightarrow$ AC $\rightarrow$ SAC $\rightarrow$ PC,
- process: submission, bidding, review, rebuttal & discussion, decision.



Relevant commands:

command	meaning
git clone url_of_repo	copy: remote $\rightarrow$ local ($1\times$), xor
git pull	copy: remote $\rightarrow$ local (grab latest version)
git add .	add all modified/new files in current directory recursively
git commit -m "comment"	save changes (locally), create checkpoint
git push	copy: local $\rightarrow$ remote

Relevant commands:

command	meaning
<code>git clone url_of_repo</code>	copy : remote $\rightarrow$ local ($1\times$), xor
<code>git pull</code>	copy : remote $\rightarrow$ local (grab latest version)
<code>git add .</code>	add all modified/new files in current directory recursively
<code>git commit -m "comment"</code>	save changes (locally), create checkpoint
<code>git push</code>	copy : local $\rightarrow$ remote

Notes

- It can be integrated with Overleaf.
- They also allow webpage hosting.

Admin

Schedule, recording, office hours, email recipe

- Schedule:
 - W1 - 5, W7 - 11: lecture + seminar; W6: reading week.
- **Recording:**
 - **No** (unless you have adjustment in place & you show up on my LfY).
 - Take **handwritten notes**, it helps learning (not passive, no multi-tasking)!
- Office hours:
 - seminars: arrange with Phil and Kaifang.
 - lecture: (i) on-the-fly (here), (ii) contact by email ($\Rightarrow$ COL 5.14).
- Emails: send ST443-related emails to me with
subject: [ST443] your topic !

Exam (100%; 120 minutes in the Spring exam period; term dates).

- understanding of the learned concepts, computing, proving,
- hint: put a paper with a pen & puzzle next to the bed.
- we use/apply: linear algebra, real analysis, probability.
- basis of marking: detailed reasoning; a systematic way

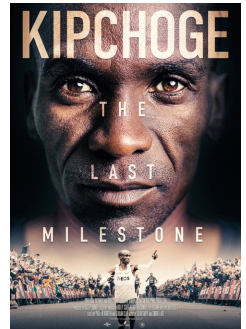
$$A \stackrel{(a)}{=} B \stackrel{(b)}{\leq} C \dots$$

where A follows from X , B holds by Y .

Admin: genAI policy

GenAI policy

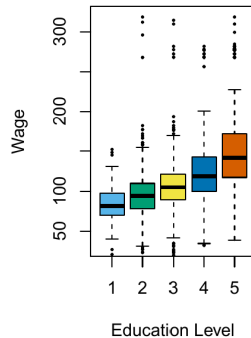
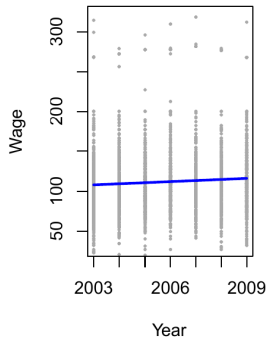
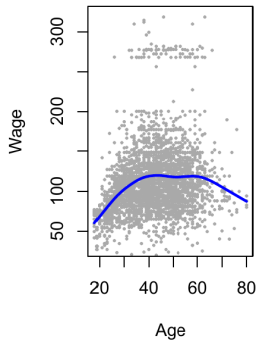
- We expect: what you submit is your work.
- Aim:
 - to demonstrate the knowledge and skills **you have learned**,
 - **to** obtain feedback for **improvement**.
- Course policy: no **GenAI**! (treated as academic misconduct)



High-level idea (@ ISLP: Ch1)

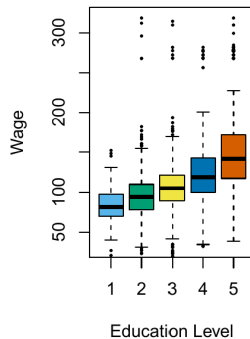
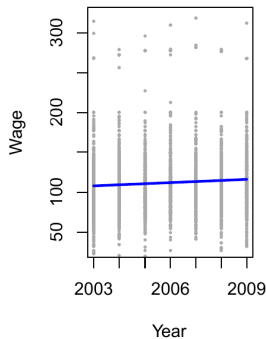
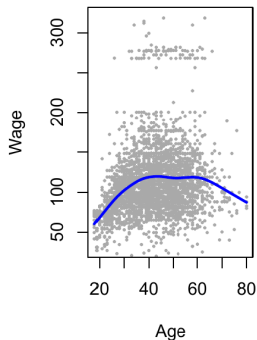
Prototype examples: task-1

Aim: understand wage vs **age**, **calendar year**, **education level** association.



Prototype examples: task-1

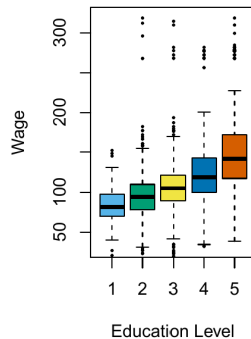
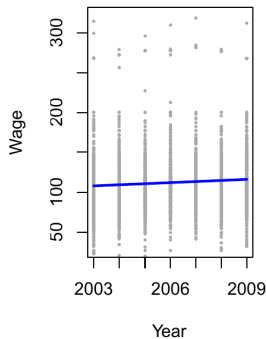
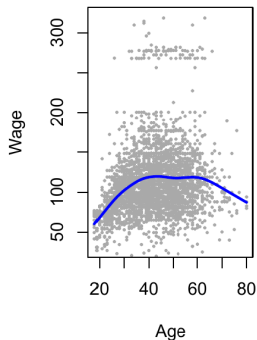
Aim: understand wage vs **age**, **calendar year**, **education level** association.



- Wage increases with
 - **age**, but decreases from ~ 60 . $\Rightarrow$ fit a curve to predict? significant variation $\Rightarrow$ other variables can come handy.

Prototype examples: task-1

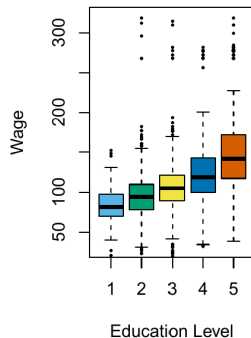
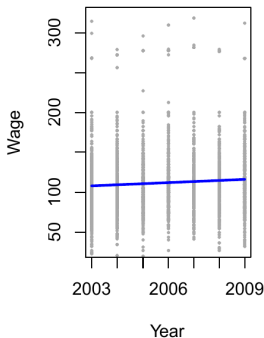
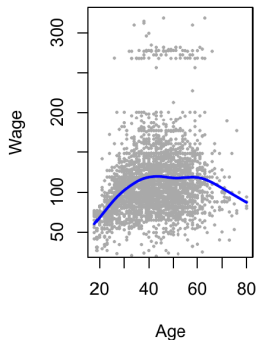
Aim: understand wage vs **age**, **calendar year**, **education level** association.



- Wage increases with
 - **age**, but decreases from ~ 60 . $\Rightarrow$ fit a curve to predict? significant variation $\Rightarrow$ other variables can come handy.
 - **year**: increases roughly linearly ($+\$10K$), grows with **education level**.

Prototype examples: task-1

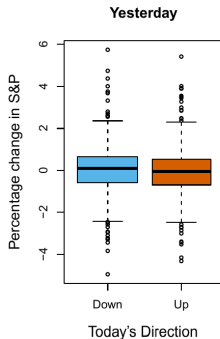
Aim: understand wage vs **age**, **calendar year**, **education level** association.



- Wage increases with
 - **age**, but decreases from ~ 60 . $\Rightarrow$ fit a curve to predict? significant variation $\Rightarrow$ other variables can come handy.
 - **year**: increases roughly linearly (+\$10K), grows with **education level**.
- Goal: capture linear & non-linear relations; **regression**.

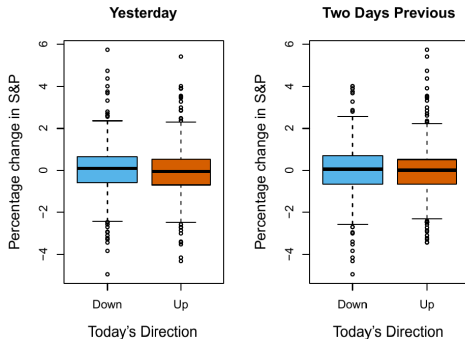
ISL: task-2

- Previous task: continuous output. Now: **categorical**/qualitative output.
- Aim: predict if **S&P 500** will go **up/down** based on historical % change.



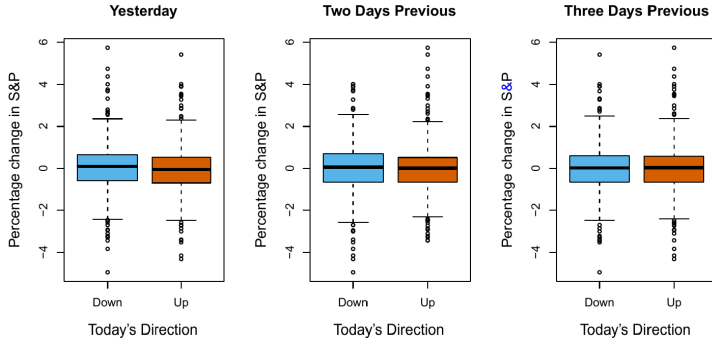
ISL: task-2

- Previous task: continuous output. Now: **categorical**/qualitative output.
- Aim: predict if **S&P 500** will go **up/down** based on historical % change.



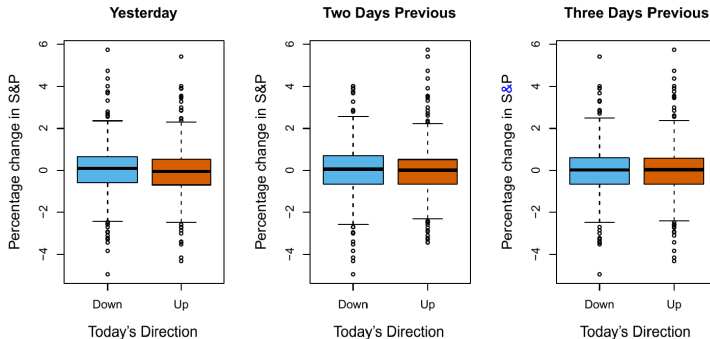
ISL: task-2

- Previous task: continuous output. Now: **categorical**/qualitative output.
- Aim: predict if **S&P 500** will go **up/down** based on historical % change.



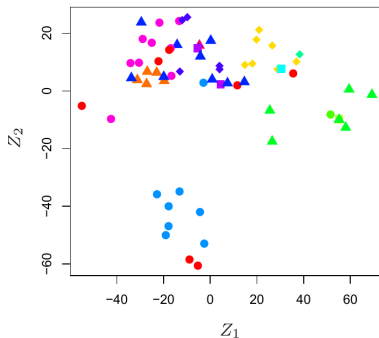
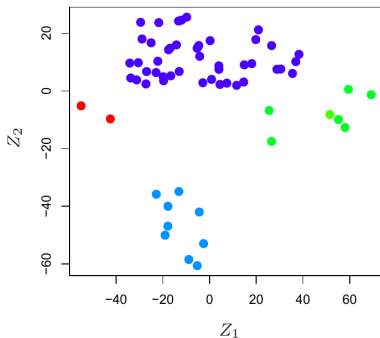
ISL: task-2

- Previous task: continuous output. Now: **categorical**/qualitative output.
- Aim: predict if **S&P 500** will go **up/down** based on historical % change.
- Hard task, but pays well.;) **classification** (ex+: spam filtering).



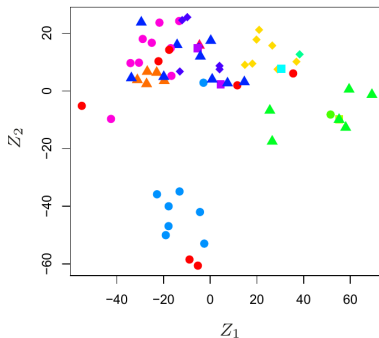
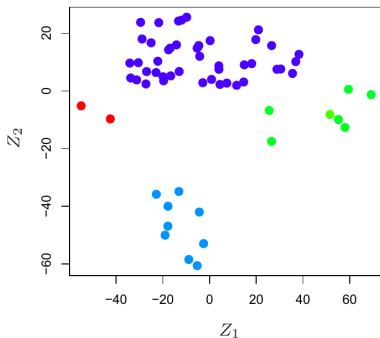
ISL: task-3

- Previous examples: input-output pairs. Now: output is missing.
- Aim:
 - marketing: categorize customers by their observed characteristics.
 - biology: gene expression categorization ($n = 6830$, $p = 40$).



ISL: task-3

- Previous examples: input-output pairs. Now: output is missing.
- Aim:
 - marketing: categorize customers by their observed characteristics.
 - biology: gene expression categorization ($n = 6830$, $p = 40$).
 - dimensionality reduction (visualization, PCA), clustering.



(Linear) algebra: recap

Typical representation: vector

$(\mathbb{R}^d, +, \cdot)$:

$$(x_1, \dots, x_d) + (y_1, \dots, y_d) := (x_1 + y_1, \dots, x_d + y_d),$$

$$\lambda \cdot (x_1, \dots, x_d) := (\lambda x_1, \dots, \lambda x_d).$$

2 operations on V : $v_1 + v_2$, λv with the 'natural' properties

Typical representation: vector

$(\mathbb{R}^d, +, \cdot)$:

$$(x_1, \dots, x_d) + (y_1, \dots, y_d) := (x_1 + y_1, \dots, x_d + y_d),$$

$$\lambda \cdot (x_1, \dots, x_d) := (\lambda x_1, \dots, \lambda x_d).$$

2 operations on V : $v_1 + v_2$, λv with the 'natural' properties

$$(v_1 + v_2) + v_3 = v_1 + (v_2 + v_3), \text{ (associativity)}$$

Typical representation: vector

$(\mathbb{R}^d, +, \cdot)$:

$$(x_1, \dots, x_d) + (y_1, \dots, y_d) := (x_1 + y_1, \dots, x_d + y_d),$$

$$\lambda \cdot (x_1, \dots, x_d) := (\lambda x_1, \dots, \lambda x_d).$$

2 operations on V : $v_1 + v_2$, λv with the 'natural' properties

$$(v_1 + v_2) + v_3 = v_1 + (v_2 + v_3), \quad (\text{associativity})$$

$$v_1 + v_2 = v_2 + v_1, \quad (\text{commutativity})$$

Typical representation: vector

$(\mathbb{R}^d, +, \cdot)$:

$$(x_1, \dots, x_d) + (y_1, \dots, y_d) := (x_1 + y_1, \dots, x_d + y_d),$$

$$\lambda \cdot (x_1, \dots, x_d) := (\lambda x_1, \dots, \lambda x_d).$$

2 operations on V : $v_1 + v_2$, λv with the 'natural' properties

$$(v_1 + v_2) + v_3 = v_1 + (v_2 + v_3), \quad (\text{associativity})$$

$$v_1 + v_2 = v_2 + v_1, \quad (\text{commutativity})$$

$$\exists 0 : v + 0 = v,$$

Typical representation: vector

$(\mathbb{R}^d, +, \cdot)$:

$$(x_1, \dots, x_d) + (y_1, \dots, y_d) := (x_1 + y_1, \dots, x_d + y_d),$$

$$\lambda \cdot (x_1, \dots, x_d) := (\lambda x_1, \dots, \lambda x_d).$$

2 operations on V : $v_1 + v_2$, λv with the 'natural' properties

$$(v_1 + v_2) + v_3 = v_1 + (v_2 + v_3), \quad (\text{associativity})$$

$$v_1 + v_2 = v_2 + v_1, \quad (\text{commutativity})$$

$$\exists 0 : v + 0 = v,$$

$$\exists -v : v + (-v) = 0,$$

Typical representation: vector

$(\mathbb{R}^d, +, \cdot)$:

$$\begin{aligned}(x_1, \dots, x_d) + (y_1, \dots, y_d) &:= (x_1 + y_1, \dots, x_d + y_d), \\ \lambda \cdot (x_1, \dots, x_d) &:= (\lambda x_1, \dots, \lambda x_d).\end{aligned}$$

2 operations on V : $v_1 + v_2$, λv with the 'natural' properties

$$(v_1 + v_2) + v_3 = v_1 + (v_2 + v_3), \quad (\text{associativity})$$

$$v_1 + v_2 = v_2 + v_1, \quad (\text{commutativity})$$

$$\exists 0 : v + 0 = v,$$

$$\exists -v : v + (-v) = 0,$$

$$a \cdot (b \cdot v) = (ab) \cdot v,$$

Typical representation: vector

$(\mathbb{R}^d, +, \cdot)$:

$$(x_1, \dots, x_d) + (y_1, \dots, y_d) := (x_1 + y_1, \dots, x_d + y_d),$$

$$\lambda \cdot (x_1, \dots, x_d) := (\lambda x_1, \dots, \lambda x_d).$$

2 operations on V : $v_1 + v_2$, λv with the 'natural' properties

$$(v_1 + v_2) + v_3 = v_1 + (v_2 + v_3), \quad (\text{associativity})$$

$$v_1 + v_2 = v_2 + v_1, \quad (\text{commutativity})$$

$$\exists 0 : v + 0 = v,$$

$$\exists -v : v + (-v) = 0,$$

$$a \cdot (b \cdot v) = (ab) \cdot v,$$

$$1 \cdot v = v,$$

Typical representation: vector

$(\mathbb{R}^d, +, \cdot)$:

$$\begin{aligned}(x_1, \dots, x_d) + (y_1, \dots, y_d) &:= (x_1 + y_1, \dots, x_d + y_d), \\ \lambda \cdot (x_1, \dots, x_d) &:= (\lambda x_1, \dots, \lambda x_d).\end{aligned}$$

2 operations on V : $v_1 + v_2$, λv with the 'natural' properties

$$(v_1 + v_2) + v_3 = v_1 + (v_2 + v_3), \quad (\text{associativity})$$

$$v_1 + v_2 = v_2 + v_1, \quad (\text{commutativity})$$

$$\exists 0 : v + 0 = v,$$

$$\exists -v : v + (-v) = 0,$$

$$a \cdot (b \cdot v) = (ab) \cdot v,$$

$$1 \cdot v = v,$$

$$a \cdot (v_1 + v_2) = a \cdot v_1 + a \cdot v_2,$$

Typical representation: vector, vectorspace (over $\mathbb{R}$)

$(\mathbb{R}^d, +, \cdot)$:

$$(x_1, \dots, x_d) + (y_1, \dots, y_d) := (x_1 + y_1, \dots, x_d + y_d),$$

$$\lambda \cdot (x_1, \dots, x_d) := (\lambda x_1, \dots, \lambda x_d).$$

2 operations on V : $v_1 + v_2$, λv with the 'natural' properties

$$(v_1 + v_2) + v_3 = v_1 + (v_2 + v_3), \quad (\text{associativity})$$

$$v_1 + v_2 = v_2 + v_1, \quad (\text{commutativity})$$

$$\exists 0 : v + 0 = v,$$

$$\exists -v : v + (-v) = 0,$$

$$a \cdot (b \cdot v) = (ab) \cdot v,$$

$$1 \cdot v = v,$$

$$a \cdot (v_1 + v_2) = a \cdot v_1 + a \cdot v_2,$$

$$(a + b) \cdot v = a \cdot v + b \cdot v \quad \text{for } \forall v_1, v_2, v_3, v \in V, a, b \in \mathbb{R}.$$

- ① Polynomials = $\mathbb{R}[x]$ (finite sums):

$$\sum_i a_i x^i + \sum_i b_i x^i := \sum_i (a_i + b_i) x^i,$$

$$\lambda \cdot \left(\sum_i a_i x^i \right) := \sum_i (\lambda a_i) x^i.$$

Vectorspace: other examples

- ① Polynomials = $\mathbb{R}[x]$ (finite sums):

$$\sum_i a_i x^i + \sum_i b_i x^i := \sum_i (a_i + b_i) x^i,$$

$$\lambda \cdot \left(\sum_i a_i x^i \right) := \sum_i (\lambda a_i) x^i.$$

- ② $\mathcal{X} \rightarrow \mathbb{R}$ -valued functions:

$$(f + g)(x) := f(x) + g(x), \quad (\lambda \cdot f)(x) := \lambda f(x).$$

Vectorspace: other examples

- ① Polynomials = $\mathbb{R}[x]$ (finite sums):

$$\sum_i a_i x^i + \sum_i b_i x^i := \sum_i (a_i + b_i) x^i,$$

$$\lambda \cdot \left(\sum_i a_i x^i \right) := \sum_i (\lambda a_i) x^i.$$

- ② $\mathcal{X} \rightarrow \mathbb{R}$ -valued functions:

$$(f + g)(x) := f(x) + g(x), \quad (\lambda \cdot f)(x) := \lambda f(x).$$

Note

1st example [with $\leq (d-1)$ -order polynomials]:

$$\mathcal{X} = [d] := \{1, \dots, d\}.$$

③ $\supset C[a, b]$ ($\mathcal{X} = [a, b]$): continuous functions.

③ $\supset C[a, b]$ ($\mathcal{X} = [a, b]$): continuous functions.

④ $(\mathbb{R}^d, \otimes, \cdot)$:

$$(x_1, \dots, x_d) \otimes (y_1, \dots, y_d) := [x_i y_j]_{i,j=1}^d$$

$$\lambda \cdot (x_1, \dots, x_d) := (\lambda x_1, \dots, \lambda x_d).$$

③ $\supset C[a, b]$ ($\mathcal{X} = [a, b]$): continuous functions.

④ $(\mathbb{R}^d, \otimes, \cdot)$: **NO!!!**

$$(x_1, \dots, x_d) \otimes (y_1, \dots, y_d) := [x_i y_j]_{i,j=1}^d \in \mathbb{R}^{d \times d} \neq \mathbb{R}^d,$$

$$\lambda \cdot (x_1, \dots, x_d) := (\lambda x_1, \dots, \lambda x_d).$$

③ $\supset C[a, b]$ ($\mathcal{X} = [a, b]$): continuous functions.

④ $(\mathbb{R}^d, \otimes, \cdot)$: NO!!!

$$(x_1, \dots, x_d) \otimes (y_1, \dots, y_d) := [x_i y_j]_{i,j=1}^d \in \mathbb{R}^{d \times d} \neq \mathbb{R}^d,$$

$$\lambda \cdot (x_1, \dots, x_d) := (\lambda x_1, \dots, \lambda x_d).$$

⑤ $(\mathbb{R}^d, \langle \cdot, \cdot \rangle, \cdot)$: similarly NO

$$\langle (x_1, \dots, x_d), (y_1, \dots, y_d) \rangle := \sum_{i=1}^d x_i y_i \in \mathbb{R} \neq \mathbb{R}^d,$$

$$\lambda \cdot (x_1, \dots, x_d) := (\lambda x_1, \dots, \lambda x_d).$$

Similarity: inner product

What is the similarity of 2 vectors?

A natural candidate: $\langle \mathbf{a}, \mathbf{b} \rangle := \sum_{i \in [d]} a_i b_i$ ($\mathbf{a}, \mathbf{b} \in \mathbb{R}^d$)

$\langle \cdot, \cdot \rangle : V \times V \rightarrow \mathbb{R}$ is called **inner product** if for $\forall \lambda_i \in \mathbb{R}, v_i, v, w \in V$

① $\langle \lambda_1 v_1 + \lambda_2 v_2, w \rangle = \lambda_1 \langle v_1, w \rangle + \lambda_2 \langle v_2, w \rangle$ (**linearity**),

Similarity: inner product

What is the similarity of 2 vectors?

A natural candidate: $\langle \mathbf{a}, \mathbf{b} \rangle := \sum_{i \in [d]} a_i b_i$ ($\mathbf{a}, \mathbf{b} \in \mathbb{R}^d$)

$\langle \cdot, \cdot \rangle : V \times V \rightarrow \mathbb{R}$ is called **inner product** if for $\forall \lambda_i \in \mathbb{R}, v_i, v, w \in V$

- ① $\langle \lambda_1 v_1 + \lambda_2 v_2, w \rangle = \lambda_1 \langle v_1, w \rangle + \lambda_2 \langle v_2, w \rangle$ (linearity),
- ② $\langle v, w \rangle = \langle w, v \rangle$ (symmetry),

Similarity: inner product

What is the similarity of 2 vectors?

A natural candidate: $\langle \mathbf{a}, \mathbf{b} \rangle := \sum_{i \in [d]} a_i b_i$ ($\mathbf{a}, \mathbf{b} \in \mathbb{R}^d$)

$\langle \cdot, \cdot \rangle : V \times V \rightarrow \mathbb{R}$ is called **inner product** if for $\forall \lambda_i \in \mathbb{R}, v_i, v, w \in V$

- ① $\langle \lambda_1 v_1 + \lambda_2 v_2, w \rangle = \lambda_1 \langle v_1, w \rangle + \lambda_2 \langle v_2, w \rangle$ (linearity),
- ② $\langle v, w \rangle = \langle w, v \rangle$ (symmetry),
- ③ $\langle v, v \rangle \geq 0; \langle v, v \rangle = 0 \Leftrightarrow v = 0$. (positive definiteness).

Similarity: inner product $\Rightarrow$ Euclidean space

- $(V, \langle \cdot, \cdot \rangle)$ is called Euclidean space.
- Examples beyond $\mathbb{R}^d$:
 - $(\mathbb{R}^{d_1 \times d_2}, \langle \mathbf{A}, \mathbf{B} \rangle_F = \text{tr}(\mathbf{A}^\top \mathbf{B}) = \sum_{i \in [d_1], j \in [d_2]} A_{i,j} B_{i,j})$.
 - $(C[a, b], \langle f, g \rangle = \int_a^b f(x)g(x)dx)$.

Why do we like inner products?

It induces a notion of

- length (norm): $\|v\| := \sqrt{\langle v, v \rangle}$.

Similarity: norm, angle

Why do we like inner products?

It induces a notion of

- length (norm): $\|v\| := \sqrt{\langle v, v \rangle}$.
- angle: CBS inequality

$$\langle v, w \rangle \leq \|v\| \|w\|.$$

Hence, for $v, w \neq 0$,

$$[-1, 1] \ni \frac{\langle v, w \rangle}{\|v\| \|w\|} =: \cos(v, w).$$

What are the properties of (the induced / a) norm?

$\|\cdot\| : V \rightarrow [0, \infty)$ is called **norm** if

- 1 $\|v\| = 0$ iff. $v = 0$ (norm separates points),

What are the properties of (the induced / a) norm?

$\|\cdot\| : V \rightarrow [0, \infty)$ is called **norm** if

- ① $\|v\| = 0$ iff. $v = 0$ (norm separates points),
- ② $\|\lambda v\| = |\lambda| \|v\| \quad \forall \lambda \in \mathbb{R}, \forall v \in V$ (positive homogeneity),

What are the properties of (the induced / a) norm?

$\|\cdot\| : V \rightarrow [0, \infty)$ is called **norm** if

- ① $\|v\| = 0$ iff. $v = 0$ (norm separates points),
- ② $\|\lambda v\| = |\lambda| \|v\| \quad \forall \lambda \in \mathbb{R}, \forall v \in V$ (positive homogeneity),
- ③ $\|v + w\| \leq \|v\| + \|w\| \quad \forall v, w \in V$ (triangle inequality).

What are the properties of (the induced / a) norm?

$\|\cdot\| : V \rightarrow [0, \infty)$ is called **norm** if

- ① $\|v\| = 0$ iff. $v = 0$ (norm separates points),
- ② $\|\lambda v\| = |\lambda| \|v\| \quad \forall \lambda \in \mathbb{R}, \forall v \in V$ (positive homogeneity),
- ③ $\|v + w\| \leq \|v\| + \|w\| \quad \forall v, w \in V$ (triangle inequality).

Note:

- norm $\Rightarrow$ metric: $\rho(f, g) = \|f - g\| \Rightarrow$
- allows one to study **continuity, convergence**.

Similarity: norm $\Rightarrow$ normed space

$(V, \|\cdot\|)$ is called **normed space**. Examples:

- $(\mathbb{R}, |\cdot|)$.

Similarity: norm $\Rightarrow$ normed space

$(V, \|\cdot\|)$ is called **normed space**. Examples:

- $(\mathbb{R}, |\cdot|)$.
- $\left(\mathbb{R}^d, \|\mathbf{x}\|_p = \left[\sum_{i \in [d]} |x_i|^p\right]^{\frac{1}{p}}\right), p \in [1, \infty)$:
 - $p = 2$: $\|\mathbf{x}\|_2 = \sqrt{\sum_{i \in [d]} x_i^2}$ (**Euclidean**),

Similarity: norm $\Rightarrow$ normed space

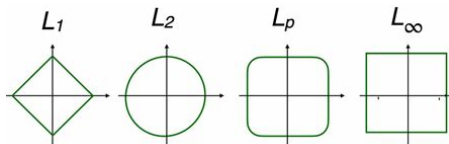
$(V, \|\cdot\|)$ is called **normed space**. Examples:

- $(\mathbb{R}, |\cdot|)$.
- $\left(\mathbb{R}^d, \|\mathbf{x}\|_p = \left[\sum_{i \in [d]} |x_i|^p\right]^{\frac{1}{p}}\right), p \in [1, \infty)$:
 - $p = 2$: $\|\mathbf{x}\|_2 = \sqrt{\sum_{i \in [d]} x_i^2}$ (Euclidean),
 - $p = 1$: $\|\mathbf{x}\|_1 = \sum_{i \in [d]} |x_i|$ (Manhattan),

Similarity: norm $\Rightarrow$ normed space

$(V, \|\cdot\|)$ is called **normed space**. Examples:

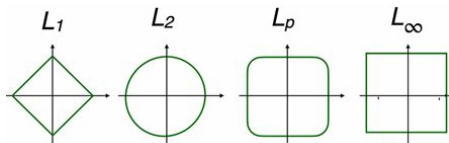
- $(\mathbb{R}, |\cdot|)$.
- $\left(\mathbb{R}^d, \|\mathbf{x}\|_p = \left[\sum_{i \in [d]} |x_i|^p\right]^{\frac{1}{p}}\right), p \in [1, \infty)$:
 - $p = 2$: $\|\mathbf{x}\|_2 = \sqrt{\sum_{i \in [d]} x_i^2}$ (Euclidean),
 - $p = 1$: $\|\mathbf{x}\|_1 = \sum_{i \in [d]} |x_i|$ (Manhattan),
 - $p = \infty$: $\|\mathbf{x}\|_\infty := \lim_{p \rightarrow \infty} \|\mathbf{x}\|_p = \max_{i \in [d]} |x_i|$ (maximum norm).



Similarity: norm $\Rightarrow$ normed space

$(V, \|\cdot\|)$ is called **normed space**. Examples:

- $(\mathbb{R}, |\cdot|)$.
- $\left(\mathbb{R}^d, \|\mathbf{x}\|_p = \left[\sum_{i \in [d]} |x_i|^p\right]^{\frac{1}{p}}\right), p \in [1, \infty)$

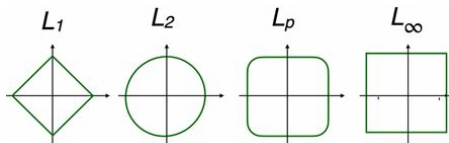


- $\left(C[a, b], \|f\|_p = \left[\int_a^b |f(x)|^p dx\right]^{\frac{1}{p}}\right), 1 \leq p < \infty.$

Similarity: norm $\Rightarrow$ normed space

$(V, \|\cdot\|)$ is called **normed space**. Examples:

- $(\mathbb{R}, |\cdot|)$.
- $(\mathbb{R}^d, \|\mathbf{x}\|_p = [\sum_{i \in [d]} |x_i|^p]^{\frac{1}{p}}), p \in [1, \infty)$



- $(C[a, b], \|f\|_p = [\int_a^b |f(x)|^p dx]^{\frac{1}{p}}), 1 \leq p < \infty.$
- $(C[a, b], \|f\|_\infty = \max_{x \in [a, b]} |f(x)|), p = \infty.$

Similarity: norm vs inner product

Relations:

- $|\langle a, b \rangle| \leq \|a\| \|b\|$ (CBS),

Similarity: norm vs inner product

Relations:

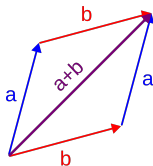
- $|\langle a, b \rangle| \leq \|a\| \|b\|$ (CBS),
- $4\langle a, b \rangle = \|a + b\|^2 - \|a - b\|^2$ (polarization identity),

Similarity: norm vs inner product

Relations:

- $|\langle a, b \rangle| \leq \|a\| \|b\|$
- $4\langle a, b \rangle = \|a + b\|^2 - \|a - b\|^2$
- $\|a + b\|^2 + \|a - b\|^2 = (*)$

(CBS),
(polarization identity),
(parallelogram rule).

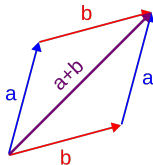


Similarity: norm vs inner product

Relations:

- $|\langle a, b \rangle| \leq \|a\| \|b\|$
- $4\langle a, b \rangle = \|a + b\|^2 - \|a - b\|^2$
- $\|a + b\|^2 + \|a - b\|^2 = (*)$

(CBS),
(polarization identity),
(parallelogram rule).



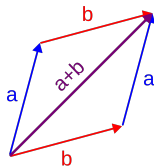
$$(*) = \langle a + b, a + b \rangle + \langle a - b, a - b \rangle$$

Similarity: norm vs inner product

Relations:

- $|\langle a, b \rangle| \leq \|a\| \|b\|$
- $4\langle a, b \rangle = \|a + b\|^2 - \|a - b\|^2$
- $\|a + b\|^2 + \|a - b\|^2 = (*)$

(CBS),
(polarization identity),
(parallelogram rule).



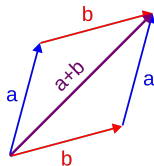
$$\begin{aligned} (*) &= \langle a + b, a + b \rangle + \langle a - b, a - b \rangle \\ &= 2 \left(\|a\|^2 + \|b\|^2 \right) \pm 2\langle a, b \rangle \end{aligned}$$

Similarity: norm vs inner product

Relations:

- $|\langle a, b \rangle| \leq \|a\| \|b\|$
- $4\langle a, b \rangle = \|a + b\|^2 - \|a - b\|^2$
- $\|a + b\|^2 + \|a - b\|^2 = (*)$

(CBS),
(polarization identity),
(parallelogram rule).



$$\begin{aligned} (*) &= \langle a + b, a + b \rangle + \langle a - b, a - b \rangle \\ &= 2 \left(\|a\|^2 + \|b\|^2 \right) \pm 2\langle a, b \rangle = 2\|a\|^2 + 2\|b\|^2. \end{aligned}$$

The parallelogram rule holds in an inner product space.

Example when the parallelogram rule fails

$C[0, 1]$ with $\|f\|_{\infty} = \max_{x \in [0, 1]} |f(x)|$:

$$f(x) := 1 - x, \quad g(x) := x$$

Example when the parallelogram rule fails

$C[0, 1]$ with $\|f\|_\infty = \max_{x \in [0, 1]} |f(x)|$:

$$f(x) := 1 - x, \quad g(x) := x,$$

$$\|f + g\|_\infty^2 + \|f - g\|_\infty^2 = \|1\|_\infty^2 + \underbrace{\|1 - 2x\|_\infty}_{\in [-1, 1]}^2 = 1^2 + 1^2 = 2,$$

Example when the parallelogram rule fails

$C[0, 1]$ with $\|f\|_\infty = \max_{x \in [0, 1]} |f(x)|$:

$$f(x) := 1 - x, \quad g(x) := x,$$

$$\|f + g\|_\infty^2 + \|f - g\|_\infty^2 = \|1\|_\infty^2 + \underbrace{\|1 - 2x\|_\infty^2}_{\in [-1, 1]} = 1^2 + 1^2 = 2,$$

$$2 \left(\|f\|_\infty^2 + \|g\|_\infty^2 \right) = 2 \underbrace{\|1 - x\|_\infty^2}_{\in [0, 1]} + 2 \underbrace{\|x\|_\infty^2}_{\in [0, 1]} = 2 + 2 = 4.$$

Example when the parallelogram rule fails

$C[0, 1]$ with $\|f\|_\infty = \max_{x \in [0, 1]} |f(x)|$:

$$f(x) := 1 - x, \quad g(x) := x,$$

$$\|f + g\|_\infty^2 + \|f - g\|_\infty^2 = \|1\|_\infty^2 + \|\underbrace{1 - 2x}_{\in [-1, 1]}\|_\infty^2 = 1^2 + 1^2 = 2,$$

$$2 \left(\|f\|_\infty^2 + \|g\|_\infty^2 \right) = 2 \|\underbrace{1 - x}_{\in [0, 1]}\|_\infty^2 + 2 \|\underbrace{x}_{\in [0, 1]}\|_\infty^2 = 2 + 2 = 4.$$

Characterization

A norm is **induced by an inner product** iff

$$\|f + g\|^2 + \|f - g\|^2 = 2 \left(\|f\|^2 + \|g\|^2 \right) \quad \forall f, g.$$

- Motivation:

$$\begin{aligned}x_1 + 2x_2 &= 5, \\ 3x_1 + 4x_2 &= 6.\end{aligned}$$

- Motivation:

$$\begin{aligned}x_1 + 2x_2 &= 5, \\ 3x_1 + 4x_2 &= 6.\end{aligned}$$

- One is lazy to write this much. $\Rightarrow$ With matrices:

$$\mathbf{Ax} = \mathbf{b}, \quad \mathbf{A} = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} 5 \\ 6 \end{bmatrix}.$$

- $\mathbf{A} + \mathbf{B}$, $\lambda \mathbf{A}$ (vector space): $\mathbf{A}, \mathbf{B} \in \mathbb{R}^{d_1 \times d_2}$, $\lambda \in \mathbb{R}$,

$$\mathbf{A} + \mathbf{B} := [A_{i,j} + B_{i,j}]_{i,j=1,1}^{d_1,d_2} \in \mathbb{R}^{d_1 \times d_2}, \quad \lambda \mathbf{A} := [\lambda A_{i,j}]_{i,j=1}^{d_1,d_2} \in \mathbb{R}^{d_1 \times d_2}.$$

- $\mathbf{A} + \mathbf{B}$, $\lambda \mathbf{A}$ (vector space): $\mathbf{A}, \mathbf{B} \in \mathbb{R}^{d_1 \times d_2}$, $\lambda \in \mathbb{R}$,

$$\mathbf{A} + \mathbf{B} := [A_{i,j} + B_{i,j}]_{i,j=1,1}^{d_1,d_2} \in \mathbb{R}^{d_1 \times d_2}, \quad \lambda \mathbf{A} := [\lambda A_{i,j}]_{i,j=1}^{d_1,d_2} \in \mathbb{R}^{d_1 \times d_2}.$$

- $\mathbf{AB}$ (ring): $\mathbf{A} \in \mathbb{R}^{d_1 \times d_2}$, $\mathbf{B} \in \mathbb{R}^{d_2 \times d_3}$, defined with inner product

$$\mathbf{AB} = [\langle A_{i,:}, B_{:,j} \rangle]_{i,j=1,1}^{d_1,d_3} \in \mathbb{R}^{d_1 \times d_3}.$$

- $\mathbf{A} \in \mathbb{R}^{d_1 \times d_2}$ is an $\mathbb{R}^{d_2} \rightarrow \mathbb{R}^{d_1}$ linear map

$$\mathbf{A}(\mathbf{v}_1 + \mathbf{v}_2) = \mathbf{A}\mathbf{v}_1 + \mathbf{A}\mathbf{v}_2, \quad \mathbf{A}(\lambda\mathbf{v}) = \lambda(\mathbf{A}\mathbf{v}).$$

- Functional analysis: One works similarly on vector/Euclidean/normed/Hilbert/Banach spaces.

Pseudo-inverse: $\exists!$, characterization

- For any $\mathbf{A} \in \mathbb{R}^{m \times n}$ there is a unique $\mathbf{A}^- \in \mathbb{R}^{n \times m}$:

$$\mathbf{A}\mathbf{A}^-\mathbf{A} = \mathbf{A},$$

Pseudo-inverse: $\exists!$, characterization

- For any $\mathbf{A} \in \mathbb{R}^{m \times n}$ there is a unique $\mathbf{A}^- \in \mathbb{R}^{n \times m}$:

$$\mathbf{A}\mathbf{A}^-\mathbf{A} = \mathbf{A},$$

$$\mathbf{A}^-\mathbf{A}\mathbf{A}^- = \mathbf{A}^-,$$

Pseudo-inverse: $\exists!$, characterization

- For any $\mathbf{A} \in \mathbb{R}^{m \times n}$ there is a unique $\mathbf{A}^- \in \mathbb{R}^{n \times m}$:

$$\mathbf{A}\mathbf{A}^-\mathbf{A} = \mathbf{A},$$

$$\mathbf{A}^-\mathbf{A}\mathbf{A}^- = \mathbf{A}^-,$$

$$(\mathbf{A}\mathbf{A}^-)^\top = \mathbf{A}\mathbf{A}^-,$$

Pseudo-inverse: $\exists!$, characterization

- For any $\mathbf{A} \in \mathbb{R}^{m \times n}$ there is a unique $\mathbf{A}^- \in \mathbb{R}^{n \times m}$:

$$\mathbf{A}\mathbf{A}^-\mathbf{A} = \mathbf{A},$$

$$\mathbf{A}^-\mathbf{A}\mathbf{A}^- = \mathbf{A}^-,$$

$$(\mathbf{A}\mathbf{A}^-)^\top = \mathbf{A}\mathbf{A}^-,$$

$$(\mathbf{A}^-\mathbf{A})^\top = (\mathbf{A}^-\mathbf{A}).$$

Pseudo-inverse: $\exists!$, characterization

- For any $\mathbf{A} \in \mathbb{R}^{m \times n}$ there is a unique $\mathbf{A}^- \in \mathbb{R}^{n \times m}$:

$$\mathbf{A}\mathbf{A}^-\mathbf{A} = \mathbf{A},$$

$$\mathbf{A}^-\mathbf{A}\mathbf{A}^- = \mathbf{A}^-,$$

$$(\mathbf{A}\mathbf{A}^-)^\top = \mathbf{A}\mathbf{A}^-,$$

$$(\mathbf{A}^-\mathbf{A})^\top = (\mathbf{A}^-\mathbf{A}).$$

Remains

Why is $\mathbf{A}^-$ useful? How to compute it?

Pseudo-inverse: usefulness – an example

Problem: solution of

$$\mathbf{Ax} = \mathbf{b}$$

might not exist or may not be unique.

Pseudo-inverse: usefulness – an example

Problem: **solution** of

$$\mathbf{Ax} = \mathbf{b}$$

might not exist or may not be unique.

The pseudo-inverse based solution $\hat{\mathbf{x}} = \mathbf{A}^+ \mathbf{b}$ handles this issue and it is 'optimal'.

- **smallest error**: for any $\mathbf{x} \in \mathbb{R}^n$

$$\|\mathbf{Ax} - \mathbf{b}\|_2 \geq \|\mathbf{A}\hat{\mathbf{x}} - \mathbf{b}\|_2.$$

Pseudo-inverse: usefulness – an example

Problem: solution of

$$\mathbf{Ax} = \mathbf{b}$$

might not exist or may not be unique.

The pseudo-inverse based solution $\hat{\mathbf{x}} = \mathbf{A}^+ \mathbf{b}$ handles this issue and it is 'optimal'.

- **smallest error**: for any $\mathbf{x} \in \mathbb{R}^n$

$$\|\mathbf{Ax} - \mathbf{b}\|_2 \geq \|\mathbf{A}\hat{\mathbf{x}} - \mathbf{b}\|_2.$$

- **smallest norm**: among the $\mathbf{x}$ vectors for which '=' holds $\hat{\mathbf{x}}$ has minimal Euclidean norm.

- Generalization of inverse: if $\mathbf{A}^{-1}$ exists, then $\mathbf{A}^{-} = \mathbf{A}^{-1}$.

Pseudo-inverse: properties

- Generalization of inverse: if $\mathbf{A}^{-1}$ exists, then $\mathbf{A}^{-} = \mathbf{A}^{-1}$.
- For $\mathbb{R}^{m \times n} \ni \mathbf{0}$: $\mathbf{0}^{-} = \mathbf{0} \in \mathbb{R}^{n \times m}$.

Pseudo-inverse: properties

- Generalization of inverse: if $\mathbf{A}^{-1}$ exists, then $\mathbf{A}^{-} = \mathbf{A}^{-1}$.
- For $\mathbb{R}^{m \times n} \ni \mathbf{0}$: $\mathbf{0}^{-} = \mathbf{0} \in \mathbb{R}^{n \times m}$.
- For $\Sigma = \text{diag}(\sigma_i) := \text{diag}\left((\sigma_i)_{i=1}^{\min(m,n)}\right) \in \mathbb{R}^{m \times n}$ diagonal:

$$\Sigma^{-} = \text{diag}(\sigma_i^{-}) \in \mathbb{R}^{n \times m},$$

$$\sigma^{-} = \begin{cases} \frac{1}{\sigma} & \text{if } \sigma \neq 0, \\ 0 & \text{if } \sigma = 0. \end{cases}$$

Pseudo-inverse: properties

- Generalization of inverse: if $\mathbf{A}^{-1}$ exists, then $\mathbf{A}^- = \mathbf{A}^{-1}$.
- For $\mathbb{R}^{m \times n} \ni \mathbf{0}$: $\mathbf{0}^- = \mathbf{0} \in \mathbb{R}^{n \times m}$.
- For $\Sigma = \text{diag}(\sigma_i) := \text{diag}\left((\sigma_i)_{i=1}^{\min(m,n)}\right) \in \mathbb{R}^{m \times n}$ diagonal:

$$\Sigma^- = \text{diag}(\sigma_i^-) \in \mathbb{R}^{n \times m},$$

$$\sigma^- = \begin{cases} \frac{1}{\sigma} & \text{if } \sigma \neq 0, \\ 0 & \text{if } \sigma = 0. \end{cases}$$

- $(\mathbf{A}^-)^- = \mathbf{A}$.

Pseudo-inverse: properties

- Generalization of inverse: if $\mathbf{A}^{-1}$ exists, then $\mathbf{A}^- = \mathbf{A}^{-1}$.
- For $\mathbb{R}^{m \times n} \ni \mathbf{0}$: $\mathbf{0}^- = \mathbf{0} \in \mathbb{R}^{n \times m}$.
- For $\Sigma = \text{diag}(\sigma_i) := \text{diag}\left((\sigma_i)_{i=1}^{\min(m,n)}\right) \in \mathbb{R}^{m \times n}$ diagonal:

$$\Sigma^- = \text{diag}(\sigma_i^-) \in \mathbb{R}^{n \times m},$$

$$\sigma^- = \begin{cases} \frac{1}{\sigma} & \text{if } \sigma \neq 0, \\ 0 & \text{if } \sigma = 0. \end{cases}$$

- $(\mathbf{A}^-)^- = \mathbf{A}$.
- It commutes with transposition: $(\mathbf{A}^\top)^- = (\mathbf{A}^-)^\top$.

Pseudo-inverse: properties

- Generalization of inverse: if $\mathbf{A}^{-1}$ exists, then $\mathbf{A}^- = \mathbf{A}^{-1}$.
- For $\mathbb{R}^{m \times n} \ni \mathbf{0}$: $\mathbf{0}^- = \mathbf{0} \in \mathbb{R}^{n \times m}$.
- For $\Sigma = \text{diag}(\sigma_i) := \text{diag}\left((\sigma_i)_{i=1}^{\min(m,n)}\right) \in \mathbb{R}^{m \times n}$ diagonal:

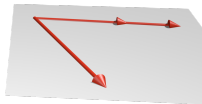
$$\Sigma^- = \text{diag}(\sigma_i^-) \in \mathbb{R}^{n \times m},$$

$$\sigma^- = \begin{cases} \frac{1}{\sigma} & \text{if } \sigma \neq 0, \\ 0 & \text{if } \sigma = 0. \end{cases}$$

- $(\mathbf{A}^-)^- = \mathbf{A}$.
- It commutes with transposition: $(\mathbf{A}^\top)^- = (\mathbf{A}^-)^\top$.
- With scalar multiplication: $(c\mathbf{A})^- = \frac{1}{c}\mathbf{A}^-$ with $c \neq 0$.

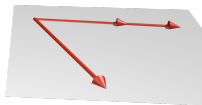
Pseudo-inverse: in case of linear independence

Linearly dependent vectors (2 examples):



Pseudo-inverse: in case of linear independence

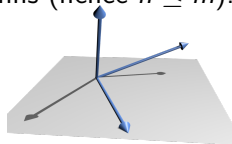
Linearly dependent vectors (2 examples):



- If $\mathbf{A} \in \mathbb{R}^{m \times n}$ has linearly independent columns (hence $n \leq m$):

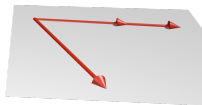
$$\mathbf{A}^- = (\mathbf{A}^\top \mathbf{A})^{-1} \mathbf{A}^\top.$$

PCA: ?



Pseudo-inverse: in case of linear independence

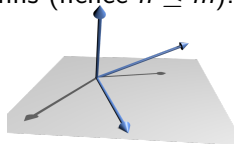
Linearly dependent vectors (2 examples):



- If $\mathbf{A} \in \mathbb{R}^{m \times n}$ has linearly independent columns (hence $n \leq m$):

$$\mathbf{A}^- = (\mathbf{A}^\top \mathbf{A})^{-1} \mathbf{A}^\top.$$

PCA: ?

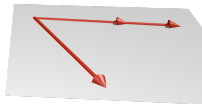


- If $\mathbf{A}$ has linearly independent rows:

$$\mathbf{A}^- = \mathbf{A}^\top (\mathbf{A} \mathbf{A}^\top)^{-1}.$$

Pseudo-inverse: in case of linear independence

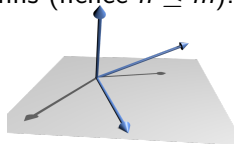
Linearly dependent vectors (2 examples):



- If $\mathbf{A} \in \mathbb{R}^{m \times n}$ has linearly independent columns (hence $n \leq m$):

$$\mathbf{A}^- = (\mathbf{A}^\top \mathbf{A})^{-1} \mathbf{A}^\top.$$

PCA: ?



- If $\mathbf{A}$ has linearly independent rows:

$$\mathbf{A}^- = \mathbf{A}^\top (\mathbf{A} \mathbf{A}^\top)^{-1}.$$

- Gaussian elimination (GE) $\Rightarrow$ checking linear (in)dependence, rank, solving linear equation, matrix inversion ($\mathbf{A} | \mathbf{I} \xrightarrow{\text{GE}} \mathbf{I} | \mathbf{A}^{-1}$).

Pseudo-inverse: computation

One can get $\mathbf{A}^-$ from the SVD of $\mathbf{A}$.

For an $\mathbf{A} \in \mathbb{R}^{m \times n}$ there is a

$$\mathbf{A} = \mathbf{U}\Sigma\mathbf{V}^\top$$

factorization (rotate $\circ$ scale $\circ$ rotate)

Pseudo-inverse: computation

One can get $\mathbf{A}^-$ from the SVD of $\mathbf{A}$.

For an $\mathbf{A} \in \mathbb{R}^{m \times n}$ there is a

$$\mathbf{A} = \mathbf{U}\Sigma\mathbf{V}^\top$$

factorization (rotate $\circ$ scale $\circ$ rotate), where

- $\mathbf{U} \in \mathbb{R}^{m \times m}$ is orthogonal: $\mathbf{U}^\top \mathbf{U} = \mathbf{I}$. (intuition? general form?
Givens rotation? optimization?)

Pseudo-inverse: computation

One can get $\mathbf{A}^-$ from the SVD of $\mathbf{A}$.

For an $\mathbf{A} \in \mathbb{R}^{m \times n}$ there is a

$$\mathbf{A} = \mathbf{U} \mathbf{\Sigma} \mathbf{V}^\top$$

factorization (rotate $\circ$ scale $\circ$ rotate), where

- $\mathbf{U} \in \mathbb{R}^{m \times m}$ is orthogonal: $\mathbf{U}^\top \mathbf{U} = \mathbf{I}$. (intuition? general form?
Givens rotation? optimization?)
- $\mathbf{V} \in \mathbb{R}^{n \times n}$ is orthogonal.

Pseudo-inverse: computation

One can get $\mathbf{A}^-$ from the SVD of $\mathbf{A}$.

For an $\mathbf{A} \in \mathbb{R}^{m \times n}$ there is a

$$\mathbf{A} = \mathbf{U} \mathbf{\Sigma} \mathbf{V}^\top$$

factorization (rotate $\circ$ scale $\circ$ rotate), where

- $\mathbf{U} \in \mathbb{R}^{m \times m}$ is orthogonal: $\mathbf{U}^\top \mathbf{U} = \mathbf{I}$. (intuition? general form?
Givens rotation? optimization?)
- $\mathbf{V} \in \mathbb{R}^{n \times n}$ is orthogonal.
- $\mathbf{\Sigma} = \text{diag}(\sigma_i) \in \mathbb{R}^{m \times n}$ is diagonal with non-negative entries, $\searrow$.

Pseudo-inverse: computation

One can get $\mathbf{A}^-$ from the SVD of $\mathbf{A}$.

For an $\mathbf{A} \in \mathbb{R}^{m \times n}$ there is a

$$\mathbf{A} = \mathbf{U} \mathbf{\Sigma} \mathbf{V}^\top$$

factorization (rotate $\circ$ scale $\circ$ rotate), where

- $\mathbf{U} \in \mathbb{R}^{m \times m}$ is orthogonal: $\mathbf{U}^\top \mathbf{U} = \mathbf{I}$. (intuition? general form?
Givens rotation? optimization?)
- $\mathbf{V} \in \mathbb{R}^{n \times n}$ is orthogonal.
- $\mathbf{\Sigma} = \text{diag}(\sigma_i) \in \mathbb{R}^{m \times n}$ is diagonal with non-negative entries, $\searrow$.
- $(\sigma_i)_{i=1}^{\min(m,n)}$: singular values.

Pseudo-inverse: computation

One can get $\mathbf{A}^-$ from the SVD of $\mathbf{A}$.

For an $\mathbf{A} \in \mathbb{R}^{m \times n}$ there is a

$$\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top$$

factorization (rotate $\circ$ scale $\circ$ rotate), where

- $\mathbf{U} \in \mathbb{R}^{m \times m}$ is orthogonal: $\mathbf{U}^\top \mathbf{U} = \mathbf{I}$. (intuition? general form?
Givens rotation? optimization?)
- $\mathbf{V} \in \mathbb{R}^{n \times n}$ is orthogonal.
- $\mathbf{\Sigma} = \text{diag}(\sigma_i) \in \mathbb{R}^{m \times n}$ is diagonal with non-negative entries, $\searrow$.
- $(\sigma_i)_{i=1}^{\min(m,n)}$: singular values.

SVD $\Rightarrow$ pseudoinverse

$$\mathbf{A}^- = \mathbf{V}\mathbf{\Sigma}^- \mathbf{U}^\top.$$

SVD reveals 'everything' about the matrix

Examples:

- $\text{rank}(\mathbf{A}) = \#$ of non-zero σ_i -s.

SVD reveals 'everything' about the matrix

Examples:

- $\text{rank}(\mathbf{A}) = \#$ of non-zero σ_i -s.
- $\|\mathbf{A}\|_F = \sqrt{\sum_i \sigma_i^2(\mathbf{A})} = \sqrt{\sum_{i,j} A_{i,j}^2} = \sqrt{\text{tr}(\mathbf{A}^\top \mathbf{A})}$: Frobenius (a.k.a. Hilbert-Schmidt) norm. Note: $\text{tr}(\mathbf{AB}) = \text{tr}(\mathbf{BA})$.

SVD reveals 'everything' about the matrix

Examples:

- $\text{rank}(\mathbf{A}) = \#$ of non-zero σ_i -s.
- $\|\mathbf{A}\|_{\mathbf{F}} = \sqrt{\sum_i \sigma_i^2(\mathbf{A})} = \sqrt{\sum_{i,j} A_{i,j}^2} = \sqrt{\text{tr}(\mathbf{A}^\top \mathbf{A})}$: Frobenius (a.k.a. Hilbert-Schmidt) norm. Note: $\text{tr}(\mathbf{AB}) = \text{tr}(\mathbf{BA})$.
- $\|\mathbf{A}\| = \sigma_1(\mathbf{A})$: operator norm.

SVD reveals 'everything' about the matrix

Examples:

- $\text{rank}(\mathbf{A}) = \#$ of non-zero σ_i -s.
- $\|\mathbf{A}\|_{\mathbf{F}} = \sqrt{\sum_i \sigma_i^2(\mathbf{A})} = \sqrt{\sum_{i,j} A_{i,j}^2} = \sqrt{\text{tr}(\mathbf{A}^\top \mathbf{A})}$: Frobenius (a.k.a. Hilbert-Schmidt) norm. Note: $\text{tr}(\mathbf{AB}) = \text{tr}(\mathbf{BA})$.
- $\|\mathbf{A}\| = \sigma_1(\mathbf{A})$: operator norm.
- Schatten p -norm:

$$\|\mathbf{A}\|_p = \|(\sigma_i(\mathbf{A}))_i\|_p, \quad p \in [1, \infty].$$

Notes:

SVD reveals 'everything' about the matrix

Examples:

- $\text{rank}(\mathbf{A}) = \#$ of non-zero σ_i -s.
- $\|\mathbf{A}\|_F = \sqrt{\sum_i \sigma_i^2(\mathbf{A})} = \sqrt{\sum_{i,j} A_{i,j}^2} = \sqrt{\text{tr}(\mathbf{A}^\top \mathbf{A})}$: Frobenius (a.k.a. Hilbert-Schmidt) norm. Note: $\text{tr}(\mathbf{AB}) = \text{tr}(\mathbf{BA})$.
- $\|\mathbf{A}\| = \sigma_1(\mathbf{A})$: operator norm.
- Schatten p -norm:

$$\|\mathbf{A}\|_p = \|(\sigma_i(\mathbf{A}))_i\|_p, \quad p \in [1, \infty].$$

Notes:

- Spec.: $p = 2$ (Frobenius), $p = \infty$ (operator norm), $p = 1$ (nuclear / trace norm; ?, Lasso: ?).

SVD reveals 'everything' about the matrix

Examples:

- $\text{rank}(\mathbf{A}) = \#$ of non-zero σ_i -s.
- $\|\mathbf{A}\|_F = \sqrt{\sum_i \sigma_i^2(\mathbf{A})} = \sqrt{\sum_{i,j} A_{i,j}^2} = \sqrt{\text{tr}(\mathbf{A}^\top \mathbf{A})}$: Frobenius (a.k.a. Hilbert-Schmidt) norm. Note: $\text{tr}(\mathbf{AB}) = \text{tr}(\mathbf{BA})$.
- $\|\mathbf{A}\| = \sigma_1(\mathbf{A})$: operator norm.
- Schatten p -norm:

$$\|\mathbf{A}\|_p = \|(\sigma_i(\mathbf{A}))_i\|_p, \quad p \in [1, \infty].$$

Notes:

- Spec.: $p = 2$ (Frobenius), $p = \infty$ (operator norm), $p = 1$ (nuclear / trace norm; $\square$), Lasso: $\square$).
- $\|\mathbf{A}\|_\infty \leq \|\mathbf{A}\|_2 \leq \|\mathbf{A}\|_1$.

Nuclear norm: intuition

- Low-rank view:

$$\mathbf{A} = \sum_i \sigma_i \mathbf{u}_i \mathbf{v}_i^\top,$$
$$\|\mathbf{A}\|_1 = \|\boldsymbol{\sigma}\|_1 = \sum_i \underbrace{|\sigma_i|}_{\sigma_i}.$$

Nuclear norm: intuition

- Low-rank view:

$$\mathbf{A} = \sum_i \sigma_i \mathbf{u}_i \mathbf{v}_i^\top,$$

$$\|\mathbf{A}\|_1 = \|\boldsymbol{\sigma}\|_1 = \sum_i \underbrace{|\sigma_i|}_{\sigma_i}.$$

- $\|\boldsymbol{\sigma}\|_1$ 'captures' the # of non-zero σ_i -s. $\Rightarrow$
- Low-rank structures (collaborative filtering):

					
A					
B					
C					
D					
E					

Low-rank approximation problem [recall: $\|\mathbf{M}\|_F^2 = \text{tr}(\mathbf{M}^\top \mathbf{M})$]

- Let $\mathbb{R}^{a \times b} \ni \mathbf{M}$.
- Task: with $k \leq r := \text{rank}(\mathbf{M})$, $\min_{\mathbf{X} \in \mathbb{R}^{a \times b} : \text{rank}(\mathbf{X})=k} \|\mathbf{X} - \mathbf{M}\|_F^2$.

Low-rank approximation problem [recall: $\|\mathbf{M}\|_F^2 = \text{tr}(\mathbf{M}^\top \mathbf{M})$]

- Let $\mathbb{R}^{a \times b} \ni \mathbf{M}$.
- Task: with $k \leq r := \text{rank}(\mathbf{M})$, $\min_{\mathbf{X} \in \mathbb{R}^{a \times b} : \text{rank}(\mathbf{X})=k} \|\mathbf{X} - \mathbf{M}\|_F^2$.
- Solution (best k -rank approximation):

$\mathbf{M} = \mathbf{U}\mathbf{D}\mathbf{V}^\top$: SVD of $\mathbf{M}$,

$\mathbf{D} = \text{diag}(\sigma_1, \dots, \sigma_r, 0, \dots, 0)$, $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_r > 0$,

$\mathbf{X}_k = \mathbf{U}\hat{\mathbf{D}}_k\mathbf{V}^\top$, $\hat{\mathbf{D}}_k = \text{diag}(\sigma_1, \dots, \sigma_k, 0, \dots, 0)$.

Solution of low-rank approximation

Proof idea

- 1 Orthogonal invariance of $\|\cdot\|_F$: for any $\mathbf{A} \in \mathbb{R}^{c \times d}$

$$\|\mathbf{U}\mathbf{A}\mathbf{V}^\top\|_F = \|\mathbf{A}\|_F, \quad \forall \mathbf{U} \in O(c), \mathbf{V} \in O(d).$$

Solution of low-rank approximation

Proof idea

- 1 Orthogonal invariance of $\|\cdot\|_F$: for any $\mathbf{A} \in \mathbb{R}^{c \times d}$

$$\|\mathbf{U}\mathbf{A}\mathbf{V}^\top\|_F = \|\mathbf{A}\|_F, \quad \forall \mathbf{U} \in O(c), \mathbf{V} \in O(d).$$

- 2 Similarly, rank is orthogonally invariant.
- 3 The problem can be reduced to the diagonal $\mathbf{D}$.

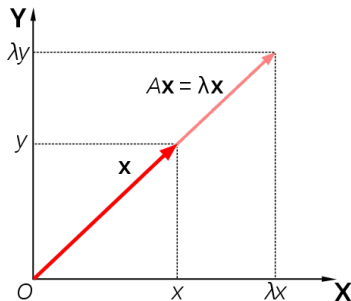
Eigensystems: eigenvectors, eigenvalues

- Simplest transformation: scaling.

Eigensystems: eigenvectors, eigenvalues

- Simplest transformation: scaling.
- $\mathbf{x} \neq \mathbf{0}$ is an eigenvector of $\mathbf{A}$ with eigenvalue $\lambda \in \mathbb{R}$ if

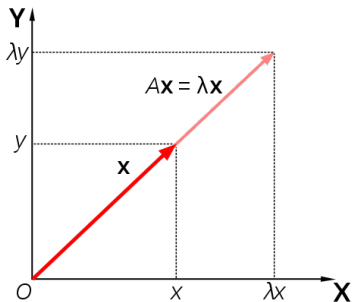
$$\mathbf{Ax} = \lambda \mathbf{x}.$$



Eigensystems: eigenvectors, eigenvalues

- Simplest transformation: scaling.
- $\mathbf{x} \neq \mathbf{0}$ is an eigenvector of $\mathbf{A}$ with eigenvalue $\lambda \in \mathbb{R}$ if

$$\mathbf{A}\mathbf{x} = \lambda\mathbf{x}.$$

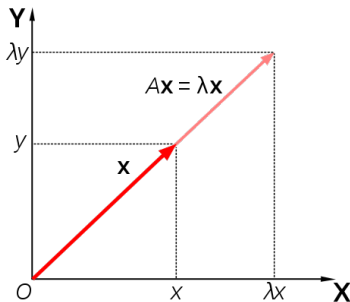


- $\mathbf{A} = \mathbf{A}^T \Rightarrow \text{all } \lambda(\mathbf{A}) \in \mathbb{R}.$

Eigensystems: eigenvectors, eigenvalues

- Simplest transformation: scaling.
- $\mathbf{x} \neq \mathbf{0}$ is an eigenvector of $\mathbf{A}$ with eigenvalue $\lambda \in \mathbb{R}$ if

$$\mathbf{Ax} = \lambda \mathbf{x}.$$



- $\mathbf{A} = \mathbf{A}^T \Rightarrow$ all $\lambda(\mathbf{A}) \in \mathbb{R}$.
- Size of $\mathbf{A}$? Eigenvalues: roots of $k_{\mathbf{A}}(\lambda) = \det(\mathbf{A} - \lambda \mathbf{I})$, ?.

A few properties with $\mathbf{A} \in \mathbb{R}^{n \times n}$:

- Formal **definition**:

$$\det(\mathbf{A}) = \sum_{\sigma=(\sigma_i)_{i=1}^n \in S_n} \text{sign}(\sigma) a_{1,\sigma_1} a_{2,\sigma_2} \dots a_{n,\sigma_n},$$

where $\text{sign}(\sigma) := (-1)^{N(\sigma)}$ is the **signature of the permutation σ** ,

$N(\sigma) = \#$ of inversions in $\sigma = \#\{i < j \text{ and } \sigma(i) > \sigma(j) : i, j \in [n]\}$.

A few properties with $\mathbf{A} \in \mathbb{R}^{n \times n}$:

- Formal **definition**:

$$\det(\mathbf{A}) = \sum_{\sigma=(\sigma_i)_{i=1}^n \in S_n} \text{sign}(\sigma) a_{1,\sigma_1} a_{2,\sigma_2} \cdots a_{n,\sigma_n},$$

where $\text{sign}(\sigma) := (-1)^{N(\sigma)}$ is the **signature of the permutation σ** ,

$N(\sigma) = \#$ of inversions in $\sigma = \#\{i < j \text{ and } \sigma(i) > \sigma(j) : i, j \in [n]\}$.

- For $n = 1, 2$: simple explicit formula,

$$\det([a]) = a, \qquad \det\left(\begin{bmatrix} a & b \\ c & d \end{bmatrix}\right) = ad - bc.$$

Determinant

A few properties with $\mathbf{A} \in \mathbb{R}^{n \times n}$:

- Formal **definition**:

$$\det(\mathbf{A}) = \sum_{\sigma=(\sigma_i)_{i=1}^n \in S_n} \text{sign}(\sigma) a_{1,\sigma_1} a_{2,\sigma_2} \cdots a_{n,\sigma_n},$$

where $\text{sign}(\sigma) := (-1)^{N(\sigma)}$ is the **signature of the permutation σ** ,

$N(\sigma) = \#$ of inversions in $\sigma = \#\{i < j \text{ and } \sigma(i) > \sigma(j) : i, j \in [n]\}$.

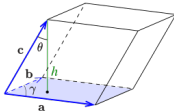
- For $n = 1, 2$: simple explicit formula,

$$\det([a]) = a, \qquad \det\left(\begin{bmatrix} a & b \\ c & d \end{bmatrix}\right) = ad - bc.$$

- Generally: recursion (**Laplace expansion**).

Determinant – continued

- Geometric intuition: (signed) volume of the parallelepiped spanned by the columns of $\mathbf{A}$ ($= [\mathbf{a}, \mathbf{b}, \mathbf{c}]$ below).



- $\det(c\mathbf{A}) = c^n \det(\mathbf{A})$, $\det(\mathbf{A}) = \det(\mathbf{A}^\top)$, $\det(\mathbf{AB}) = \det(\mathbf{A}) \det(\mathbf{B})$.
- For $\mathbf{A}$ triangular: $\det(\mathbf{A}) = \prod_{i \in [n]} A_{i,i}$, $[n] := \{1, 2, \dots, n\}$.
- Relation to eigenvalues: $\det(\mathbf{A}) = \prod_{i \in [n]} \lambda_i(\mathbf{A})$, $\lambda_i(\mathbf{A}) \in \mathbb{C}$.

Eigensystems: continued

Examples:

- Identity: $\mathbf{A} = \mathbf{I}$.

Eigensystems: continued

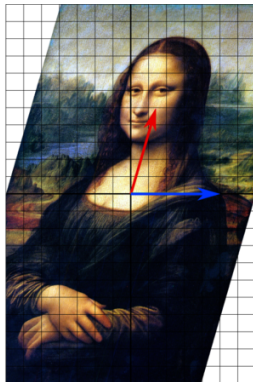
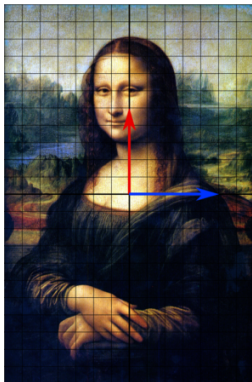
Examples:

- Identity: $\mathbf{A} = \mathbf{I}$.
- Diagonal matrix: $\mathbf{A} = \text{diag}(a_i)$, spec.: reflection.

Eigensystems: continued

Examples:

- Identity: $\mathbf{A} = \mathbf{I}$.
- Diagonal matrix: $\mathbf{A} = \text{diag}(a_i)$, spec.: reflection.
- Shear mapping on Mona Lisa:



- Diagonal matrix: the eigensystem can be chosen to be orthogonal – as we saw it.

- Diagonal matrix: the eigensystem can be chosen to be orthogonal – as we saw it.
- A symmetric $\mathbf{A}$ ($\mathbf{A} = \mathbf{A}^\top$) behaves similarly: $\exists$

$$\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{U}^\top,$$

where $\mathbf{\Sigma} = \text{diag}(\lambda_i)$, $\mathbf{U}$: orthogonal.

Symmetric matrices are nice

- Diagonal matrix: the eigensystem can be chosen to be orthogonal – as we saw it.
- A symmetric $\mathbf{A}$ ($\mathbf{A} = \mathbf{A}^\top$) behaves similarly: $\exists$

$$\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{U}^\top,$$

where $\mathbf{\Sigma} = \text{diag}(\lambda_i)$, $\mathbf{U}$: orthogonal.

An application of eigensystems

- principal component analysis (PCA): [demo now](#).

PCA example: 100%

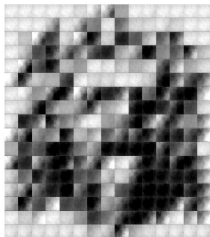


(A)

PCA example: 100% $\rightarrow$ 1%



(A)

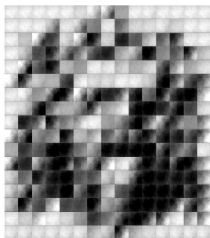


(B)

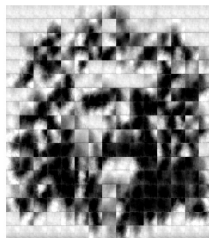
PCA example: 100% $\rightarrow$ 2%



(A)



(B)

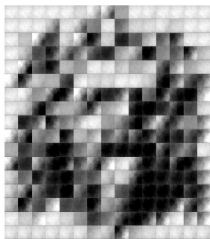


(C)

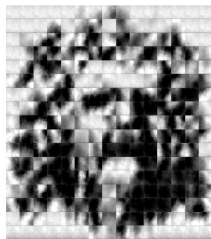
PCA example: 100% $\rightarrow$ 5%



(A)



(B)



(C)

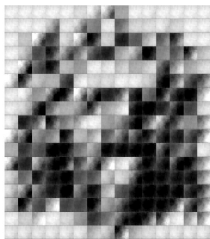


(D)

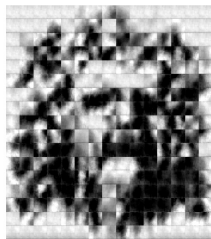
PCA example: 100% $\rightarrow$ 10%



(A)



(B)



(C)



(D)

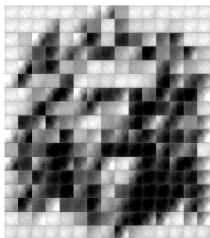


(E)

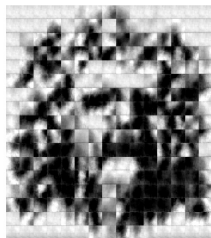
PCA example: 100% $\rightarrow$ 20%



(A)



(B)



(C)



(D)



(E)



(F)

Note: right vs. left eigenvector

Sizes: $\mathbf{A} \in \mathbb{R}^{n \times n}$, $\mathbf{x} \in \mathbb{R}^n$ (column vector).

Eigenvector := right eigenvector

$$\mathbf{Ax} = \lambda \mathbf{x}.$$

Note: right vs. left eigenvector

Sizes: $\mathbf{A} \in \mathbb{R}^{n \times n}$, $\mathbf{x} \in \mathbb{R}^n$ (column vector), $\mathbf{y} \in \mathbb{R}^{1 \times n}$ (row vector).

Eigenvector := right eigenvector

$$\mathbf{A}\mathbf{x} = \lambda\mathbf{x}.$$

Left eigenvector

$$\mathbf{y}\mathbf{A} = \kappa\mathbf{y}.$$

Note: right vs. left eigenvector

Sizes: $\mathbf{A} \in \mathbb{R}^{n \times n}$, $\mathbf{x} \in \mathbb{R}^n$ (column vector), $\mathbf{y} \in \mathbb{R}^{1 \times n}$ (row vector).

Eigenvector := right eigenvector

$$\mathbf{A}\mathbf{x} = \lambda\mathbf{x}.$$

Left eigenvector

$$\mathbf{y}\mathbf{A} = \kappa\mathbf{y}. \quad (\Leftrightarrow \mathbf{A}^\top \mathbf{y}^\top = \kappa \mathbf{y}^\top)$$

In words:

- The left eigenvectors of $\mathbf{A}$ are the right eigenvectors of $\mathbf{A}^\top$ (modulo $\top$).
- The left and right eigenvalues of $\mathbf{A}$ are identical $\Leftarrow$

$$k_{\mathbf{A}}(\lambda) = \det(\mathbf{A} - \lambda \mathbf{I}_n) = \det((\mathbf{A} - \lambda \mathbf{I}_n)^\top) = \det(\mathbf{A}^\top - \lambda \mathbf{I}_n) = k_{\mathbf{A}^\top}(\lambda).$$

Warmup:

- team, references & their authors, ML hints,
- admin,
- prototype tasks.
- linear algebra: recap.

Local summary

Warmup:

- team, references & their authors, ML hints,
- admin,
- prototype tasks.
- linear algebra: recap.

